Rockchip Android Multi Audio 开发指南

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前言

概述

本文主要介绍Android Multi Audio，开发或技术支持人员通过阅读此文档，对Android平台Mulit Audio模块有一个初步的了解，帮助阅读者开发和调试。

产品版本

芯片名称	内核版本	Android版本
适配所有芯片	适配所有版本	12/13/14/15/16

读者对象

本文档（本指南）主要适用于以下工程师：

技术支持工程师

软件开发工程师

修订记录

版本号	作者	修改日期	修改说明
V1.0.0	刘兴亮	2022-04-28	初始版本
V1.0.1	刘兴亮	2022-10-20	添加新增
V1.0.2	刘兴亮	2023-05-11	添加多声卡输出原理图
V1.0.3	刘兴亮	2025-02-05	标准化代码、接口更新
V1.1.0	刘兴亮	2026-01-19	更新原理图、新增MPDemo、更新无序列表显示

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1. 背景介绍

Android系统多个AudioTrack同时运行时默认混音，系统会按照AudioPolicy服务中预先设置的策略选择优先级最高的设备，混音后的数据会从此设备对应的声卡输出。如果硬件上支持多个声卡，如何让多个声卡同时输出相同声音或者同时分别输出不同的声音?

如果设备硬件上支持多个同种类型的声卡（比如：HDMI0/HDMI1），Android系统如何区分?

HDMIIN和MIC需要同时录音，如何支持?

新增设备类型导致GMS认证Audio相关case会fail，如何避免?

...

针对上述问题，本文整理并开发了一套MultiAudio代码，用户可依据开发需求自由配置，且不影响GMS认证，帮助用户充分利用硬件多声卡，提升产品性能。

2. RK MultiAudio功能说明

支持的功能

- 同种类型设备（例如：HDMI0/HDMI1） 插拔识别
- 声音分离（hdmi0/hdmi1/dp0/dp1 能够同时分别播放不同的音频数据）
- 多声卡同时输出相同声音
- 不同设备同时录音（mic/hdmiin能够同时分别录音）
- 提供新增设备类型方法（符合谷歌标准）
- 提供新增扩展库的方法（代码复用，不用重新开发新的hal）

2.1 同种类型设备插拔识别及扩展

2.1.1 Android插拔检测

以hdmi为例，Android系统插拔检测是通过响应extcon下的节点变化

```
sys/class/extcon/extcon0/state
```

```
#其中：HDMI=0表示HDMI未插入 HDMI=1表示HDMI插入
rk3588_t:/ # cat sys/class/extcon/extcon12/state
HDMI=0
rk3588_t:/ # cat sys/class/extcon/extcon13/state
HDMI=0
```

上面extcon12和extcon13对应的state都是HDMI=0或者HDMI=1，Android系统只能知道当前有HDMI设备发生插拔动作，无法区分是HDMI0还是HDMI1，因此需要设法区分。

2.1.2 同类型设备区分

以hdmi0/hdmi1为例（其它设备可参考），通过如下节点判断

```
sys/class/extcon/extcon*/device/uevent
```

dts中需要需配置alias

```
diff --git a/arch/arm64/boot/dts/rockchip/rk3588.dtsi
b/arch/arm64/boot/dts/rockchip/rk3588.dtsi
index b40eee3eddc9..90242e13645a 100644
--- a/arch/arm64/boot/dts/rockchip/rk3588.dtsi
+++ b/arch/arm64/boot/dts/rockchip/rk3588.dtsi
@@ -15,6 +15,8 @@
                ethernet0 = &gmac0;
                hdptx0 = &hdptxphy0;
                hdptx1 = &hdptxphy1;
+               hdmi0 = &hdmi0;
+               hdmi1 = &hdmi1;
        };
```

配置后查看对应节点情况如下：

```
rk3588_t:/ # cat sys/class/extcon/extcon12/device/uevent
DRIVER=dwhdmi-rockchip
OF_NAME=hdmi
OF_FULLNAME=/hdmi@fde80000
OF_COMPATIBLE_0=rockchip,rk3588-dw-hdmi
OF_COMPATIBLE_N=1
OF_ALIAS_0=hdmi0
MODALIAS=of:NhdmiT(null)Crockchip,rk3588-dw-hdmi
rk3588_t:/ # cat sys/class/extcon/extcon13/device/uevent
DRIVER=dwhdmi-rockchip
OF_NAME=hdmi
OF_FULLNAME=/hdmi@fdea0000
OF_COMPATIBLE_0=rockchip,rk3588-dw-hdmi
OF_COMPATIBLE_N=1
OF_ALIAS_0=hdmi1
MODALIAS=of:NhdmiT(null)Crockchip,rk3588-dw-hdmi
```

其中OF_ALIAS_0对应的值可以区分hdmi0/hdmi1。

2.1.3 框架修改

框架修改见frameworks如下修改（Android 13）

```
commit 62e285277c574cc001bb212c86b39ad25e7522ba
Author: shine.liu <shine.liu@rock-chips.com>
Date:   Wed Apr 27 10:48:57 2022 +0800

    Audio: add multiaudio support for tablet

    1. add plug in/out uevent support for hdmi/dp/hdmirx
    2. set and save volumes of multi devices at the same time

    Signed-off-by: shine.liu <shine.liu@rock-chips.com>
    Change-Id: I7c09273791fe95c8853c70b9cd22cace424cc28b
    (cherry picked from commit 14d6ef289ca6b8e6c6d80b25cd0601351ea1260d)

 media/java/android/media/AudioManager.java                      | 13 +++++++++++
 media/java/android/media/AudioSystem.java                       |  6 ++++++
 services/core/java/com/android/server/ExtconStateObserver.java  | 63
++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++--
 services/core/java/com/android/server/ExtconUEventObserver.java |  5 +++++
 services/core/java/com/android/server/WiredAccessoryManager.java | 37
++++++++++++++++++++++++++++++++++---
 services/core/java/com/android/server/audio/AudioService.java   | 62
++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++---
 6 files changed, 178 insertions(+), 8 deletions(-)
```

```
commit d8bdbfc57e9ba2d8eb5623c3057ce4c7abf7e3b6 (tag: android-13.0-mid-rkr4)
Author: shine.liu <shine.liu@rock-chips.com>
Date:   Thu Jan 5 17:24:39 2023 +0800

    MultiAudio: Add VX_ROCKCHIP_XXX to audio framework

    Signed-off-by: shine.liu <shine.liu@rock-chips.com>
    Change-Id: I9819637692562960 47e3f0202d436df12f791f0d

 media/libaudioclient/AidlConversion.cpp                    | 15 +++++++++++++++
 services/audiopolicy/enginedefault/src/Engine.cpp          |  6 +++---
 services/audiopolicy/managerdefault/AudioPolicyManager.cpp | 22 ++++++++++++++++------
 3 files changed, 29 insertions(+), 14 deletions(-)
```

2.1.4 selinux问题

如果实际有开selinux，需要添加sys/class/extcon/extcon*/device/uevent权限，如下：

device/rockchip/rk3588/sepolicy_vendor

```
commit ae28d4463c5c290471423690cf23a45fd1396f8b (HEAD)
Author: shine.liu <shine.liu@rock-chips.com>
Date:   Fri Mar 31 09:44:19 2023 +0800

    sepolicy_vendor: add hdmi/dp/hdmirx uevent nodes for plug.

    Signed-off-by: shine.liu <shine.liu@rock-chips.com>
    Change-Id: Iba380828fef0a31a8da4f5b6e056058b86636a70

diff --git a/sepolicy_vendor/genfs_contexts b/sepolicy_vendor/genfs_contexts
index cfc37bb..0842c68 100644
--- a/sepolicy_vendor/genfs_contexts
+++ b/sepolicy_vendor/genfs_contexts
@@ -19,6 +19,7 @@ genfscon sysfs /devices/platform/fdc30000.rkvdec-ccu/wakeup
u:object_r:sysfs
 genfscon sysfs /devices/platform/fdc38100.rkvdec-core/wakeup    u:object_r:sysfs_wakeup:s0
 genfscon sysfs /devices/platform/fdc48100.rkvdec-core/wakeup    u:object_r:sysfs_wakeup:s0
 genfscon sysfs /devices/platform/fde50000.dp/extcon     u:object_r:sysfs_extcon:s0
+genfscon sysfs /devices/platform/fde60000.dp/extcon     u:object_r:sysfs_extcon:s0
 genfscon sysfs /devices/platform/fdec0000.edp/extcon    u:object_r:sysfs_extcon:s0
 genfscon sysfs /devices/platform/fe190000.pcie/wakeup   u:object_r:sysfs_wakeup:s0
 genfscon sysfs /devices/platform/fe2e0000.mmc/mmc_host/mmc0     u:object_r:sysfs_mmc:s0
@@ -34,3 +35,6 @@ genfscon sysfs
 genfscon sysfs /devices/platform/usbdrd3_0/fc000000.usb/wakeup  u:object_r:sysfs_wakeup:s0
+genfscon sysfs /devices/platform/fde80000.hdmi/uevent   u:object_r:sysfs_extcon:s0
+genfscon sysfs /devices/platform/fdea0000.hdmi/uevent   u:object_r:sysfs_extcon:s0
+genfscon sysfs /devices/platform/fdee0000.hdmirx-controller/uevent
u:object_r:sysfs_extcon:s0
```

2.1.5 多个speaker如何区分

上面介绍过拔插设备的区分实现，对与speaker这种非拔插设备该如何区分并分别使用？

实际碰到的多speaker机型硬件不支持hdmi/spdif声卡功能，可以复用现有的hdmi/spdif对应的代码，减少开发工作量。假设三个speaker分别是speaker0、speaker1和speaker2，可以将speaker1当作hdmi，将speaker2当作spdif，具体修改步骤如下：

- 由于hdmi/spdif是可拔插设备，首先要将hdmi/spdif输出设置为attachedDevices，从而能够被系统识别到。

```
diff --git a/audio_policy_configuration_multihal.xml
b/audio_policy_configuration_multihal.xml
index b3ec1db5..f0e01198 100644
--- a/audio_policy_configuration_multihal.xml
+++ b/audio_policy_configuration_multihal.xml
@@ -77,6 +77,7 @@
         <module name="ext_1" halVersion="2.0">
```

```
             <attachedDevices>
                 <item>HDMIIn</item>
+                <item>HDMI Out</item>
             </attachedDevices>
             <mixPorts>
                 <mixPort name="hdmi_default_output" role="source">
@@ -139,6 +140,9 @@

         <!--3 extend Audio HAL-->
         <module name="ext_3" halVersion="2.0">
+            <attachedDevices>
+                <item>SPDIF Out</item>
+            </attachedDevices>
             <mixPorts>
                 <mixPort name="spdif_default_output" role="source">
                     <profile name="" format="AUDIO_FORMAT_PCM_16_BIT"
```

- 在hal中将speaker1的实际声卡名称添加到HDMI_OUT_NAME，将speaker2的实际声卡名称添加到SPDIF_OUT_NAME，三个speaker驱动注册的声卡名称尽量区分开，以免hal中识别紊乱。

以上是最简单的方法，实际当然也可以通过address来区分，这个就需要从上改到下，如果框架不熟悉，难度较大。

2.2 声音分离

Android系统来自多个AudioTrack播放的数据默认会在AudioFlinger中进行混音最后输出到优先级最高的设备。多屏异显情况下，不同的屏幕上用不同的app去播放不同的视频，默认也是混音。声音分离就是将不同屏幕上播放视频的声音数据不混音，分别送到不同的声卡中去。

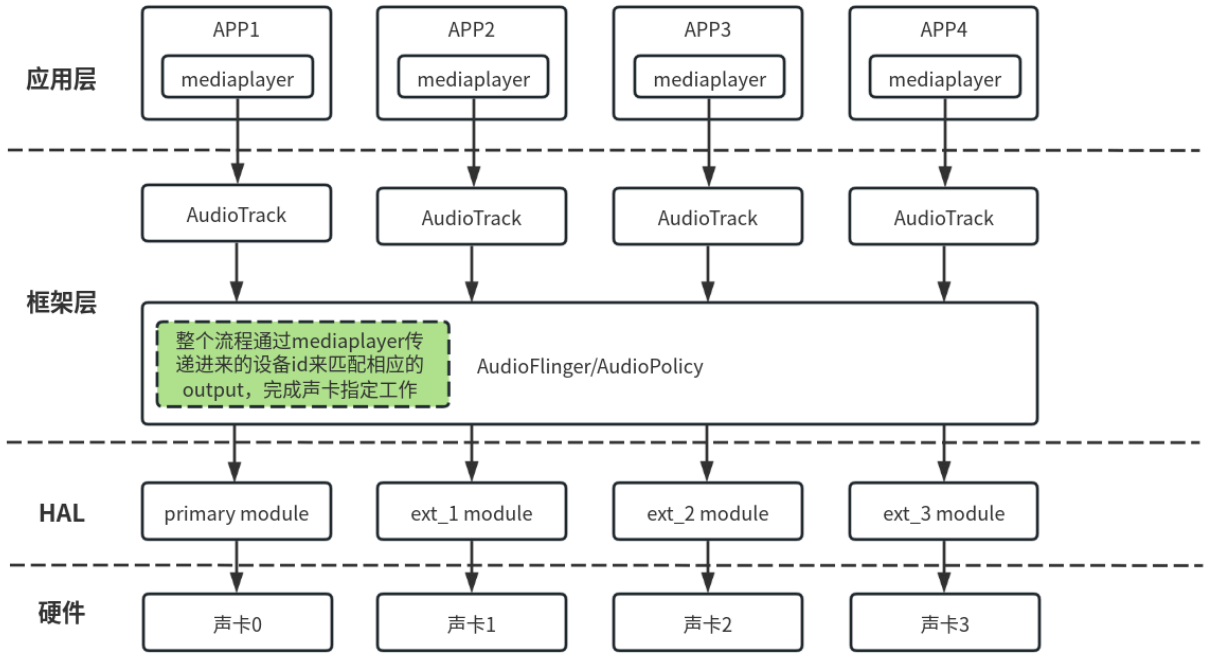
2.2.1 宏开关

```
diff --git a/BoardConfig.mk b/BoardConfig.mk
index 07f7801e..93ca005e 100644
--- a/BoardConfig.mk
+++ b/BoardConfig.mk
@@ -378,7 +378,7 @@ BOARD_USER_FAKETOUCH ?= true
 endif

 # disable BOARD_SUPPORT_MULTIAUDIO default
-BOARD_SUPPORT_MULTIAUDIO ?= false
+BOARD_SUPPORT_MULTIAUDIO ?= true

 #for Camera autofocus support
 CAMERA_SUPPORT_AUTOFOCUS ?= false
```

2.2.2 原理图



2.2.3 接口及调用方法

2.2.3.1 java标准接口

用户自己开发播放应用可以调用MediaPlayer的接口setPreferredDevice来选择具体的device

```
    /**
     * Specifies an audio device (via an {@link AudioDeviceInfo} object) to route
     * the output from this MediaPlayer.
     * @param deviceInfo The {@link AudioDeviceInfo} specifying the audio sink or source.
     *  If deviceInfo is null, default routing is restored.
     * @return true if succesful, false if the specified {@link AudioDeviceInfo} is non-null
and
     * does not correspond to a valid audio device.
     */
    @Override
    public boolean setPreferredDevice(AudioDeviceInfo deviceInfo)
```

接口使用方法(以选择speaker为例)

```
import android.media.AudioDeviceInfo;
import android.media.AudioManager;
import android.media.MediaPlayer;

AudioDeviceInfo[] mOutputDevices;
AudioManager mAudioManager;
```

```
mAudioManager = (AudioManager)context.getSystemService(Context.AUDIO_SERVICE);
MediaPlayer mediaPlayer = new MediaPlayer();
mediaPlayer.setDataSource(xxx);
mOutputDevices = mAudioManager.getDevices(AudioManager.GET_DEVICES_OUTPUTS);
for (AudioDeviceInfo device : mOutputDevices) {
    if (device.getType() == AudioDeviceInfo.TYPE_BUILTIN_SPEAKER)
        mediaPlayer.setPreferredDevice(device);
}
mediaPlayer.prepare();
```

各种设备对应的type

```
hdmi0 ----------- AudioDeviceInfo.TYPE_HDMI
hdmi1 ----------- AudioDeviceInfo.TYPE_VX_HDMI0
dp0 ------------- AudioDeviceInfo.TYPE_LINE_DIGITAL
dp1 ------------- AudioDeviceInfo.TYPE_VX_SPDIF0
```

2.2.3.2 native层通过包名指定声卡

如果用户不自己开发播放应用，且测试的第三方应用不开源，可通过这种方式指定声卡。

native层有一个demo，实际可参考demo进行开发。测试效果可将MultiAudioTest置1。

```
diff --git a/media/libaudioclient/include/media/AudioTrack.h
b/media/libaudioclient/include/media/AudioTrack.h
index d05398b3c7..5a3270624d 100644
--- a/media/libaudioclient/include/media/AudioTrack.h
+++ b/media/libaudioclient/include/media/AudioTrack.h
@@ -35,7 +35,7 @@
 #include "android/media/IAudioTrack.h"
 #include "android/media/IAudioTrackCallback.h"

-#define MultiAudioTest 0
+#define MultiAudioTest 1
 namespace android {

 using content::AttributionSourceState;
```

具体实现如下：

```
diff --git a/media/libaudioclient/AudioTrack.cpp b/media/libaudioclient/AudioTrack.cpp
index c7e7f0f255..976ae10467 100644
--- a/media/libaudioclient/AudioTrack.cpp
+++ b/media/libaudioclient/AudioTrack.cpp
@@ -572,6 +571,56 @@ status_t AudioTrack::set(
     } else {
         mPackageName = String16("default");
     }
+    String8 tmp = String8(mPackageName);
```

```
+    unsigned int numPorts;
+    unsigned int generation1;
+    unsigned int generation;
+    struct audio_port_v7 *audioPorts = nullptr;
+    int attempts = 10;
+    do {
+        if (attempts-- < 0) {
+            free(audioPorts);
+            ALOGE("Query audio ports time out");
+            break;
+        }
+        numPorts = 0;
+        status = AudioSystem::listAudioPorts(
+                AUDIO_PORT_ROLE_SINK, AUDIO_PORT_TYPE_DEVICE, &numPorts, nullptr,
+                &generation1);
+        if (numPorts == 0 || status != NO_ERROR) {
+            free(audioPorts);
+            ALOGE("Number of audio ports should not be zero");
+            break;
+        }
+        audioPorts = (struct audio_port_v7 *)realloc(
+                    audioPorts, numPorts * sizeof(struct audio_port_v7));
+        status = AudioSystem::listAudioPorts(
+                AUDIO_PORT_ROLE_SINK, AUDIO_PORT_TYPE_DEVICE, &numPorts, audioPorts,
+                &generation);
+        if (status != NO_ERROR) {
+            free(audioPorts);
+            ALOGE("Query audio ports failed");
+        }
+    } while (generation1 != generation);
+    for (int i = 0 ; i < numPorts; i++) {
+        ALOGD("listDeviceType: %d type:0x%x addr:%s id:%d", i,
+            audioPorts[i].ext.device.type, audioPorts[i].ext.device.address,
audioPorts[i].id);
+        if (strstr(audioPorts[i].ext.device.address, "speaker0") != nullptr
+            && audioPorts[i].ext.device.type == AUDIO_DEVICE_OUT_SPEAKER
+            && strstr(tmp.string(), "RockVideoPlayer")) {
+           if (streamType == AUDIO_STREAM_MUSIC) {
+                selectedDeviceId = audioPorts[i].id;
+                ALOGD("select id:%d", selectedDeviceId);
+           }
+        } else if (audioPorts[i].ext.device.type == AUDIO_DEVICE_OUT_HDMI
+                        && strstr(tmp.string(), "gallery3d") != nullptr) {
+           if (streamType == AUDIO_STREAM_MUSIC) {
+                selectedDeviceId = audioPorts[i].id;
+                ALOGD("select id:%d", selectedDeviceId);
+           }
+        }
+    }
+    free(audioPorts);
 #endif
 #endif
```

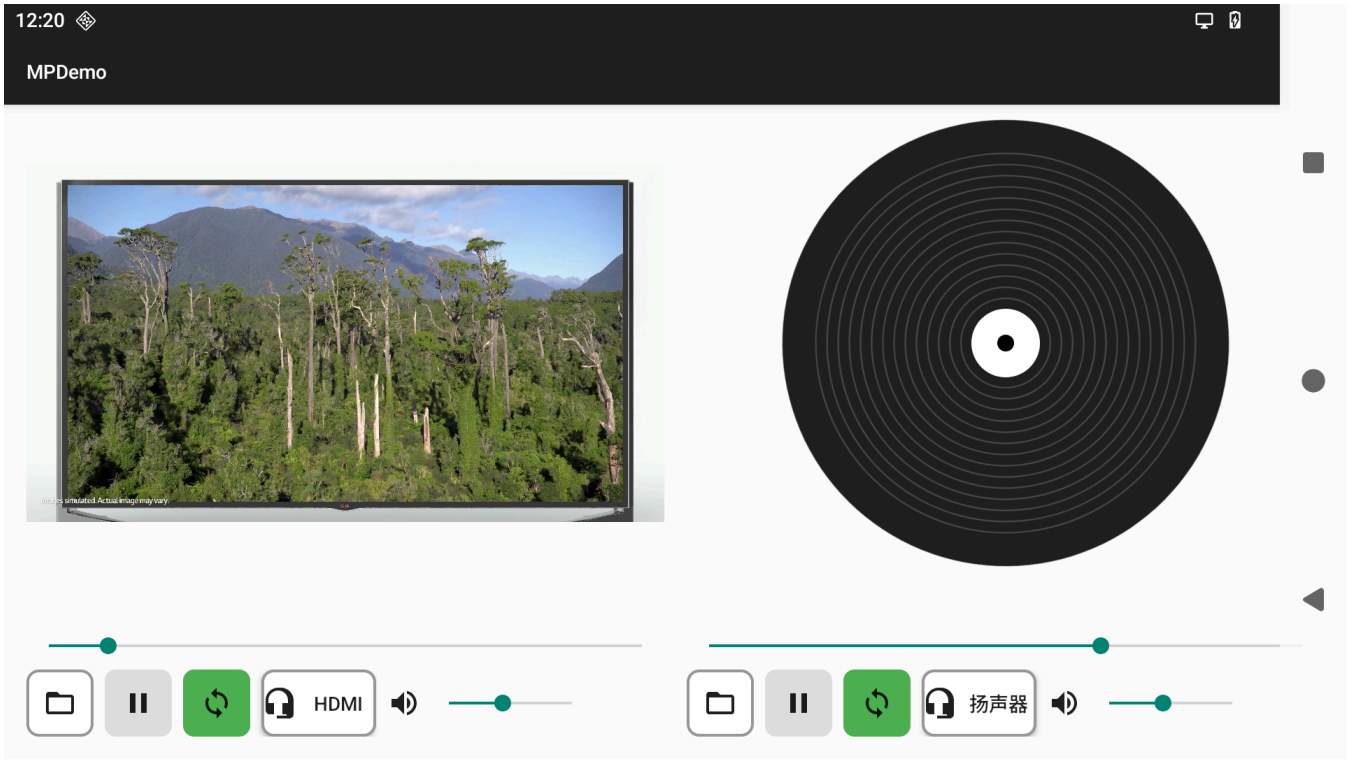
```
     // Note mPortId is not valid until the track is created, so omit mPortId in ALOG for
set.
@@ -1806,6 +1855,12 @@ audio_io_handle_t AudioTrack::getOutput() const

 status_t AudioTrack::setOutputDevice(audio_port_handle_t deviceId) {
     AutoMutex lock(mLock);
+#if SUPPORT_MULTIAUDIO
+#if MultiAudioTest
+    if (deviceId == AUDIO_PORT_HANDLE_NONE)
+        return NO_ERROR;
+#endif
+#endif
     ALOGV("%s(%d): deviceId=%d mSelectedDeviceId=%d",
             __func__, mPortId, deviceId, mSelectedDeviceId);
     if (mSelectedDeviceId != deviceId) {
```

2.2.4 MPDemo

MPDemo是一个Android多媒体演示应用程序，支持双路独立播放器，可以分别播放音频和视频文件。该应用特别设计用于测试和演示多媒体播放功能，包括输出设备选择、循环播放、播放控制等功能。

界面如下:



用户如果开发异声应用，可以参考此应用，源码下载地址:

```
https://github.com/wsgwx/MPDemo.git
```

循环播放多次后，出现视频倍速、音频卡顿的现象，可通过如下修改解决:

```
diff --git a/services/audiopolicy/managerdefault/AudioPolicyManager.cpp
b/services/audiopolicy/managerdefault/AudioPolicyManager.cpp
index ab505774ae..6c5b8e6d4c 100644
--- a/services/audiopolicy/managerdefault/AudioPolicyManager.cpp
+++ b/services/audiopolicy/managerdefault/AudioPolicyManager.cpp
@@ -2860,7 +2860,7 @@ status_t AudioPolicyManager::stopSource(const
sp<SwAudioOutputDescriptor>& outpu
         }
         bool forceDeviceUpdate = false;
         if (client->hasPreferredDevice(true) &&
-                outputDesc->sameExclusivePreferredDevicesCount() < 2) {
+                outputDesc->sameExclusivePreferredDevicesCount() == 1) {
             checkStrategyRoute(client->strategy(), AUDIO_IO_HANDLE_NONE);
             forceDeviceUpdate = true;
         }
```

如果框架已包含hdmi0/hdmi1的区分修改，MPDemo源码可以加上如下修改：

```
diff --git a/src/main/java/com/example/mpdemo/MainActivity.java
b/src/main/java/com/example/mpdemo/MainActivity.java
index a30b23e..0949566 100644
--- a/src/main/java/com/example/mpdemo/MainActivity.java
+++ b/src/main/java/com/example/mpdemo/MainActivity.java
@@ -412,6 +412,8 @@ public class MainActivity extends AppCompatActivity {
             return getString(R.string.device_bluetooth);
         } else if (deviceInfo.getType() == AudioDeviceInfo.TYPE_HDMI) {
             return getString(R.string.device_hdmi);
+        } else if (deviceInfo.getType() == AudioDeviceInfo.TYPE_VX_HDMI0) {
+            return getString(R.string.device_hdmi_1);
         } else if (deviceInfo.getType() == AudioDeviceInfo.TYPE_USB_HEADSET ||
                    deviceInfo.getType() == AudioDeviceInfo.TYPE_USB_DEVICE) {
             return getString(R.string.device_usb);
diff --git a/src/main/res/values/strings.xml b/src/main/res/values/strings.xml
index 223cfe1..985bb40 100644
--- a/src/main/res/values/strings.xml
+++ b/src/main/res/values/strings.xml
@@ -26,6 +26,7 @@
     <string name="device_speaker">扬声器</string>
     <string name="device_headphone">耳机</string>
     <string name="device_bluetooth">蓝牙</string>
-    <string name="device_hdmi">HDMI</string>
+    <string name="device_hdmi">HDMI0</string>
+    <string name="device_hdmi_1">HDMI1</string>
     <string name="device_usb">USB</string>
 </resources>
```

2.3 多声卡同时输出相同声音

如果开机后播放视频，希望所有声卡都出相同声音，可通过如下方式进行配置(以3588 evb1为例)。

2.3.1 关闭宏

```
#device/rockchip/common/BoardConfig.mk
BOARD_SUPPORT_MULTIAUDIO ?= false
```

2.3.2 修改audio_policy_configuration.xml

Android12可以按照这个修改，Android13/14/15已经包含

device/rockchip/common

```
diff --git a/audio_policy_configuration.xml b/audio_policy_configuration.xml
index f0a5d1d..c8c55a6 100644
--- a/audio_policy_configuration.xml
+++ b/audio_policy_configuration.xml
@@ -82,8 +82,12 @@
                 </devicePort>
                 <devicePort tagName="HDMI Out" type="AUDIO_DEVICE_OUT_AUX_DIGITAL"
role="sink">
                 </devicePort>
+                <devicePort tagName="HDMI_1 Out" type="VX_ROCKCHIP_OUT_HDMI0" role="sink">
+                </devicePort>
                 <devicePort tagName="SPDIF Out" type="AUDIO_DEVICE_OUT_SPDIF" role="sink">
                 </devicePort>
+                <devicePort tagName="SPDIF_1 Out" type="VX_ROCKCHIP_OUT_SPDIF0"
role="sink">
+                </devicePort>

                 <devicePort tagName="Built-In Mic" type="AUDIO_DEVICE_IN_BUILTIN_MIC"
role="source">
                 </devicePort>
@@ -109,8 +113,12 @@
                       sources="primary output"/>
                 <route type="mix" sink="HDMI Out"
                       sources="primary output,hdmi_bitstream"/>
+                <route type="mix" sink="HDMI_1 Out"
+                      sources="primary output"/>
                 <route type="mix" sink="SPDIF Out"
                       sources="primary output,spdif_passthrough"/>
+                <route type="mix" sink="SPDIF_1 Out"
+                      sources="primary output"/>
                 <route type="mix" sink="primary input"
                       sources="Built-In Mic,Wired Headset Mic,BT SCO Headset Mic,HDMIIn"/>
             </routes>
```

2.3.3 框架修改（仅供参考，依据开发需求增、删）

```
diff --git a/services/audiopolicy/enginedefault/src/Engine.cpp
b/services/audiopolicy/enginedefault/src/Engine.cpp
index fd1fd420b8..87a6c920e4 100644
--- a/services/audiopolicy/enginedefault/src/Engine.cpp
+++ b/services/audiopolicy/enginedefault/src/Engine.cpp
@@ -24,6 +24,11 @@
 #define ALOGVV(a...) do { } while(0)
 #endif

+//-----------------------rk code----------
+// open all the devices simultaneously in one hal
+#define OPEN_ALL_DEVICES_SIMULT 0
+//----------------------------------------
+
 #include "Engine.h"
 #include <android-base/macros.h>
 #include <AudioPolicyManagerObserver.h>
@@ -500,6 +505,23 @@ DeviceVector Engine::getDevicesForStrategyInt(legacy_strategy strategy,
         // STRATEGY_ENFORCED_AUDIBLE, AUDIO_DEVICE_NONE otherwise
         devices.add(devices2);

+//-----------------------rk code----------
+#if OPEN_ALL_DEVICES_SIMULT
+        DeviceVector devicesOpenSimult;
+        DeviceVector devicesAlwaysOpen;
+        if (strategy == STRATEGY_MEDIA) {
+            devicesOpenSimult = availableOutputDevices.getDevicesFromTypes({
+                    AUDIO_DEVICE_OUT_SPEAKER, AUDIO_DEVICE_OUT_HDMI,
+                    VX_ROCKCHIP_OUT_HDMI0, VX_ROCKCHIP_OUT_SPDIF0,});
+            devicesAlwaysOpen = availableOutputDevices.getDevicesFromType(
+                    AUDIO_DEVICE_OUT_SPDIF);
+        }
+        if (getLastRemovableMediaDevices().size() == 0)
+            devices.add(devicesOpenSimult);
+        devices.add(devicesAlwaysOpen);
+#endif
+//----------------------------------------
+
```

2.3.4 同声策略下浏览器播放视频声音卡顿

原因：AAudio默认不支持多声卡同声

分析：多声卡同时输出会反复触发AAudio里面相关的callback函数，导致播放的音频数据断断续续

补丁：

```
diff --git a/media/libaaudio/src/legacy/AudioStreamLegacy.cpp
b/media/libaaudio/src/legacy/AudioStreamLegacy.cpp
index dd11169d8a..7ad20326e6 100644
--- a/media/libaaudio/src/legacy/AudioStreamLegacy.cpp
+++ b/media/libaaudio/src/legacy/AudioStreamLegacy.cpp
@@ -266,6 +266,8 @@ void AudioStreamLegacy::onAudioDeviceUpdate(audio_io_handle_t /* audioIo
*/,
     // get called after the stream has been deleted so log before we
     // get called after the stream has been deleted so log before we
     // touch the stream object.
     ALOGD("%s(deviceId = %d)", __func__, (int)deviceId);
+    if (deviceId == AAUDIO_UNSPECIFIED)
+        return;
     if (getDeviceId() != AAUDIO_UNSPECIFIED
             && getDeviceId() != deviceId
             && getState() != AAUDIO_STREAM_STATE_DISCONNECTED
diff --git a/media/libaaudio/src/legacy/AudioStreamTrack.cpp
b/media/libaaudio/src/legacy/AudioStreamTrack.cpp
index 6f1dc928f4..5cd8da9b5f 100644
--- a/media/libaaudio/src/legacy/AudioStreamTrack.cpp
+++ b/media/libaaudio/src/legacy/AudioStreamTrack.cpp
@@ -94,6 +94,9 @@ aaudio_result_t AudioStreamTrack::open(const AudioStreamBuilder& builder)
     }

     size_t frameCount = (size_t)builder.getBufferCapacity();
+    if (AudioTrack::getMinFrameCount(&frameCount, AUDIO_STREAM_MUSIC, getSampleRate()) !=
NO_ERROR) {
+        ALOGD("can not get frameCount");
+    }

     // To avoid glitching, let AudioFlinger pick the optimal burst size.
     int32_t notificationFrames = 0;
@@ -199,7 +202,7 @@ aaudio_result_t AudioStreamTrack::open(const AudioStreamBuilder&
builder)
     setFormat(mAudioTrack->format());
     setDeviceFormat(mAudioTrack->format());
     setSampleRate(mAudioTrack->getSampleRate());
-    setBufferCapacity(getBufferCapacityFromDevice());
+    setBufferCapacity((frameCount == 0) ? getBufferCapacityFromDevice() : frameCount);
     setFramesPerBurst(getFramesPerBurstFromDevice());
```

2.4 多声卡同时录音

比较常见的场景是mic和hdmiin同时录音，这种场景打开宏就支持，不用别的修改

```
diff --git a/BoardConfig.mk b/BoardConfig.mk
index 07f7801e..93ca005e 100644
--- a/BoardConfig.mk
+++ b/BoardConfig.mk
@@ -378,7 +378,7 @@ BOARD_USER_FAKETOUCH ?= true
 endif

 # disable BOARD_SUPPORT_MULTIAUDIO default
-BOARD_SUPPORT_MULTIAUDIO ?= false
+BOARD_SUPPORT_MULTIAUDIO ?= true

 #for Camera autofocus support
 CAMERA_SUPPORT_AUTOFOCUS ?= false
```

应用端通过AudioRecord录音通过AudioSource无法指定hdmiin，建议使用setPreferredDevice接口，如果hdmiin录音这边是自行开发，可参考如下代码：

```
        AudioDeviceInfo[] deviceList =
mAudioManager.getDevices(AudioManager.GET_DEVICES_INPUTS);
        AudioDeviceInfo preferDevice = null;

        //retry to find hdmiin, timeout: 1s
        int retry = 0;
        do {
            SystemClock.sleep(50);
            for (AudioDeviceInfo device : deviceList) {
                if (device.getInternalType() == AudioManager.DEVICE_IN_HDMI) {
                    Log.d(TAG, "find hdmiin, retry: " + retry);
                    preferDevice = device;
                    break;
                }
            }
            retry++;
            deviceList = mAudioManager.getDevices(AudioManager.GET_DEVICES_INPUTS);
        } while (preferDevice == null && retry < 20);

        mRecorder = new AudioRecord.Builder()
                        .setAudioFormat(new AudioFormat.Builder()
                                .setSampleRate(mSampleRate)
                                .setChannelMask(mChannelCount)
                                .setEncoding(mAudioEncoding)
                                .build())
                        .setAudioSource(MediaRecorder.AudioSource.DEFAULT)
                        .setBufferSizeInBytes(bufSize)
                        .build();
        if (preferDevice != null) {
            mRecorder.setPreferredDevice(preferDevice);
        } else {
            Log.d(TAG, "Did not find hdmiin, use default!, retry: " + retry);
        }
```

2.5 兼容GMS测试

GMS测试会涉及到GSI（Google System.img），GSI测试时由于框架层修改的缺失，测试会报错，可添加如下修改解决

frameworks/av

```
commit 71aee93c0787d6b4c2563f2b134ab3cb83672da5
Author: shine.liu <shine.liu@rock-chips.com>
Date:   Sat Sep 14 16:35:31 2024 +0800

    Audio: Load audiopolicy xml by priority

    Type: Fix
    Redmine ID: N/A
    Associated modifications:
        https://10.10.10.29/c/android/platform/system/media/+/235212
    Test: N/A

    Change-Id: I4271708136286ba5096a73cdc4a5c9c5b7d0e0cf
    Signed-off-by: shine.liu <shine.liu@rock-chips.com>

diff --git a/services/audiopolicy/common/managerdefinitions/src/AudioPolicyConfig.cpp
b/services/audiopolicy/common/managerdefinitions/src/AudioPolicyConfig.cpp
index 4edd11f935..29e7f6a7a4 100644
--- a/services/audiopolicy/common/managerdefinitions/src/AudioPolicyConfig.cpp
+++ b/services/audiopolicy/common/managerdefinitions/src/AudioPolicyConfig.cpp
@@ -197,8 +197,15 @@ sp<const AudioPolicyConfig>
AudioPolicyConfig::loadFromApmAidlConfigWithFallback
 // static
 sp<const AudioPolicyConfig> AudioPolicyConfig::loadFromApmXmlConfigWithFallback(
         const std::string& xmlFilePath) {
-    const std::string filePath =
-            xmlFilePath.empty() ? audio_get_audio_policy_config_file() : xmlFilePath;
+//-----------------------rk code----------
+    std::string path_single = audio_find_readable_configuration_file(
+                                "audio_policy_configuration_singlehal.xml");
+    std::string path_multi = audio_find_readable_configuration_file(
+                                "audio_policy_configuration_multihal.xml");
+    std::string path_tablet = path_multi.empty() ? path_single : path_multi;
+    const std::string filePath = path_tablet.empty() ?
+            (xmlFilePath.empty() ? audio_get_audio_policy_config_file() : xmlFilePath) :
path_tablet;
+//----------------------------------------
     auto config = sp<AudioPolicyConfig>::make();
     if (status_t status = config->loadFromXml(filePath, false /*forVts*/); status ==
NO_ERROR) {
         return config;
```

3. 调试

3.1 查看声卡

```
cat proc/asound/cards
```

```
rk3588_s:/ # cat proc/asound/cards
 0 [rockchipes8388 ]: rockchip-es8388 - rockchip-es8388
                      rockchip-es8388
 1 [rockchiphdmiin ]: rockchip_hdmiin - rockchip,hdmiin
                      rockchip,hdmiin
 2 [rockchiphdmi0  ]: rockchip-hdmi0 - rockchip-hdmi0
                      rockchip-hdmi0
 3 [rockchiphdmi1  ]: rockchip-hdmi1 - rockchip-hdmi1
                      rockchip-hdmi1
```

3.2 测试驱动是否正常

如果要测试3.1中3588的hdmi0声卡，找一个wav文件，测试命令如下：

```
tinyplay xxx.wav -D 2 -d 0
```

如果声音正常，说明hdmi0声卡驱动正常。

3.3 录音或播放测试卡住

通过debuggerd查看堆栈

```
debuggerd -b $(pidof audioserver)
debuggerd -b $(pidof system_server)
```

3.4 dump audio信息

```
dumpsys audio
dumpsys media.audio_flinger
dumpsys media.audio_policy
```

4. Bug及解决思路

4.1 更新到最新的代码后，HDMI声音没输出，没更新前是正常的

- 问题分析

```
rk3588_s:/ # cat /proc/asound/cards
0 [realtekrt5616co]: realtek_rt5616- - realtek,rt5616-codec
realtek,rt5616-codec
1 [rockchiphdmiin ]: rockchip_hdmiin - rockchip,hdmiin
rockchip,hdmiin
2 [rockchiphdmi1 ]: rockchip-hdmi1 - rockchip-hdmi1
rockchip-hdmi1
```

新的框架支持多hdmi声卡识别，如果系统只有一路hdmi，上层会识别成hdmi0, 最终会调用hdmi_hal，但hdmi_hal中不包含rockchiphdmi1，找不到声卡，因此没有声音

- 解决办法

```
将dts中rockchip,card-name = "rockchip-hdmi1"改成rockchip,card-name = "rockchip-hdmi0"
```

4.2 uboot开启了logo后, hdmi就没有声音了, 要插拔下hdmi才会有声音

- 问题分析

extcon状态更新不及时

```
rk3588_s:/ # cat /sys/class/extcon/extcon12/state
cat /sys/class/extcon/extcon12/state
HDMI=0
rk3588_s:/ # cat /sys/class/extcon/extcon11/state
cat /sys/class/extcon/extcon11/state
HDMI=0
```

- 解决办法

```
+++ b/drivers/gpu/drm/bridge/synopsys/dw-hdmi-qp.c
@@ -1716,6 +1716,14 @@ dw_hdmi_connector_detect(struct drm_connector *connector, bool force)
                else
                        result = connector_status_disconnected;
        }
+    mutex_lock(&hdmi->mutex);
+    if (result != hdmi->last_connector_result) {
+        dev_dbg(hdmi->dev, "read_hpd result: %d", result);
+        extcon_set_state_sync(hdmi->extcon, EXTCON_DISP_HDMI, result ==
connector_status_connected);
+        handle_plugged_change(hdmi,result == connector_status_connected);
+        hdmi->last_connector_result = result;
+    }
+    mutex_unlock(&hdmi->mutex);
```

4.3 SPDIF和其它声卡同时出声声音断断续续

- 解决办法

kernel包含以下三个提交

```
commit cbce2baa6fea4ca806442699a348418a909e7ec3
Author: Sugar Zhang <sugar.zhang@rock-chips.com>
Date:   Wed May 16 15:31:40 2018 +0800

    ALSA: pcm_dmaengine: always get stream position from DMA driver

    This patch fixup that the wrong position when dma desc status
    is DONE. even if the desc status is DONE, it is still able to
    get the position from the dma driver. so, just remove the judgement.

    Change-Id: I40e92bae09a002f4f5f0b2fab8b0e99fd3ee269d
    Signed-off-by: Sugar Zhang <sugar.zhang@rock-chips.com>
```

```
commit 4039352c7d055abc701b8c890d12af69bf239a2b
Author: Sugar Zhang <sugar.zhang@rock-chips.com>
Date:   Sat Mar 26 18:01:21 2022 +0800

    dmaengine: pl330: Fix unbalanced runtime PM

    This driver use runtime PM autosuspend mechanism to manager clk.
```

```
commit 78de4e20d0c97393171d033ad137bc823faf7618
Author: Sugar Zhang <sugar.zhang@rock-chips.com>
Date:   Wed Apr 27 09:38:06 2022 +0800

    dmaengine: pl330: Improve dma cyclic transfers efficiency

    Currently, the driver implements cyclic transfers by desc list.
    Each desc describes one period and started by CPU after last
    one done(tasklet), which maybe delayed due to schedule or
    heavy system load, which will cause device FIFO xrun.、
```

声卡dts配置需包含simple-audio-card,mclk-fs = <128>

```
  spdif_tx0_sound: spdif-tx0-sound {
      status = "disabled";
      compatible = "simple-audio-card";
      simple-audio-card,mclk-fs = <128>;
      simple-audio-card,name = "rockchip,spdif-tx0";
      simple-audio-card,cpu {
          sound-dai = <&spdif_tx0>;
      };
      simple-audio-card,codec {
          sound-dai = <&spdif_tx0_dc>;
      };
  };
```

4.4 hdmiin偶尔出现audio服务崩溃

- 原因分析

TimeCheck超时

- 解决办法

kernel加上如下两个修改

```
diff --git a/sound/soc/rockchip/rockchip_i2s_tdm.c b/sound/soc/rockchip/rockchip_i2s_tdm.c
index b7ffaaa7153e..e7a58d9c8c43 100644
--- a/sound/soc/rockchip/rockchip_i2s_tdm.c
+++ b/sound/soc/rockchip/rockchip_i2s_tdm.c
@@ -1173,6 +1173,10 @@ static int rockchip_i2s_tdm_trigger(struct snd_pcm_substream
*substream,
          break;
      }
+     if(substream->stream == SNDRV_PCM_STREAM_CAPTURE) {
+         substream->wait_time = msecs_to_jiffies(500);
+     }
+
      return ret;
  }
```

```
diff --git a/sound/soc/rockchip/rockchip_i2s.c b/sound/soc/rockchip/rockchip_i2s.c
index d29ab5c33468..10a8c83f4e8c 100644
--- a/sound/soc/rockchip/rockchip_i2s.c
+++ b/sound/soc/rockchip/rockchip_i2s.c
@@ -435,6 +435,9 @@ static int rockchip_i2s_trigger(struct snd_pcm_substream *substream,
                break;
        }
+       if(substream->stream == SNDRV_PCM_STREAM_CAPTURE) {
+               substream->wait_time = msecs_to_jiffies(500);
+       }
        return ret;
 }
```

蓝牙超时

```
diff --git a/audio_a2dp_hw/src/audio_a2dp_hw.cc b/audio_a2dp_hw/src/audio_a2dp_hw.cc
index 0387e4510..49e2c3689 100644
--- a/audio_a2dp_hw/src/audio_a2dp_hw.cc
+++ b/audio_a2dp_hw/src/audio_a2dp_hw.cc
@@ -54,10 +54,10 @@
  *  Constants & Macros
  ******************************************************************************/

-#define CTRL_CHAN_RETRY_COUNT 3
+#define CTRL_CHAN_RETRY_COUNT 2
 #define USEC_PER_SEC 1000000L
 #define SOCK_SEND_TIMEOUT_MS 2000 /* Timeout for sending */
-#define SOCK_RECV_TIMEOUT_MS 5000 /* Timeout for receiving */
+#define SOCK_RECV_TIMEOUT_MS 2000 /* Timeout for receiving */
 #define SEC_TO_MS 1000
 #define SEC_TO_NS 1000000000
 #define MS_TO_NS 1000000
```

4.5 多声卡同时输出其中一个卡顿

- 解决办法

pcm_dmaengine: always get stream position from DMA

This patch fixup that the wrong position when dma desc status
is DONE. even if the desc status is DONE, it is still able to
get the position from the dma driver. so, just remove the judgement.

BACKGROUND
In the Multi-Stream audio situation, Some Stream XRUNs unexpected,
These stream drived by the synchronous clk, So, it should not be.

```
diff --git a/drivers/dma/pl330.c b/drivers/dma/pl330.c
index 91592809d496..bd0ed77f7d55 100644
--- a/drivers/dma/pl330.c
+++ b/drivers/dma/pl330.c
@@ -2674,7 +2674,7 @@ pl330_tx_status(struct dma_chan *chan, dma_cookie_t cookie,
 out:
       dma_set_residue(txstate, residual);

-       return ret;
+       return DMA_IN_PROGRESS;
 }
```

4.6 同声策略下，开机前接入DP-OUT，主设备静音后再解除静音，实际音量相比静音前明显变大

- 问题分析

同声策略下，开机过程中，音量保存发生在applyAllVolumes之后，导致mIndexMap没有更新

- 解决补丁

```
diff --git a/services/core/java/com/android/server/audio/AudioService.java
b/services/core/java/com/android/server/audio/AudioService.java
index 2bde6a93f648..b796fdaaa207 100644
--- a/services/core/java/com/android/server/audio/AudioService.java
+++ b/services/core/java/com/android/server/audio/AudioService.java
@@ -8705,6 +8705,8 @@ public class AudioService extends IAudioService.Stub
                             index = (mIndexMax + 5)/10;
                         } else if (device == AudioSystem.DEVICE_OUT_HEARING_AID) {
                             index = (mIndexMax + 5)/10;
+                        } else if (isTablet() && isSyncAjustVolumeDevice(device) &&
mStreamType == AudioSystem.STREAM_MUSIC) {
+                            index = (getIndex(AudioSystem.DEVICE_OUT_SPEAKER) + 5)/10;
                         } else {
                             index = (mIndexMap.valueAt(i) + 5)/10;
                         }
```

5. 其它扩展

5.1 AudioPatch

现有的tvinput是通过AudioStream来完成hdmiin声音的录音和播放，实际上TvInputHardwareManager.java中已经实现了AudioPatch，可以通过如下补丁将AudioPatch利用起来（Android 14适用）。

- 将AudioStream屏蔽

```
diff --git a/services/core/java/com/android/server/tv/TvInputHardwareManager.java
b/services/core/java/com/android/server/tv/TvInputHardwareManager.java
index bad215e36c91..564cee0d46cf 100644
--- a/services/core/java/com/android/server/tv/TvInputHardwareManager.java
+++ b/services/core/java/com/android/server/tv/TvInputHardwareManager.java
@@ -128,7 +128,7 @@ class TvInputHardwareManager implements TvInputHal.Callback {
         mContext = context;
         mListener = listener;
         mAudioManager = (AudioManager) context.getSystemService(Context.AUDIO_SERVICE);
-        mAudioPreviewThread = new AudioPreviewThread(context);
+        //mAudioPreviewThread = new AudioPreviewThread(context);
         mHal.init();
```

- 设置tv_input中的even对应的audio_type

```
diff --git a/tv_input.cpp b/tv_input.cpp
index d42cb35..d6356d5 100644
--- a/tv_input.cpp
+++ b/tv_input.cpp
@@ -138,7 +138,7 @@ V4L2EventCallBack hinDevEventCallback(int event_type) {
     ALOGE("%s width:%d,height:%d,format:0x%x,%d",
__FUNCTION__,s_HinDevStreamWidth,s_HinDevStreamHeight,s_HinDevStreamFormat,isHdmiIn);
     event.base_event.device_info.device_id = SOURCE_HDMI1;
     event.base_event.device_info.type = TV_INPUT_TYPE_HDMI;
-    event.base_event.device_info.audio_type = AUDIO_DEVICE_NONE;
+    event.base_event.device_info.audio_type = AUDIO_DEVICE_IN_HDMI;
     event.base_event.device_info.audio_address = NULL;
     if(event.base_event.type > 0)
       s_TvInputPriv->callback->notify_ext(nullptr, &event, nullptr);
@@ -195,7 +195,7 @@ static int hin_dev_open(int deviceId, int type)
 static int generateEvent(tv_input_private_t *priv, tv_input_source_t source_type, int
event_type) {
     tv_input_event_ext_t event;
     event.base_event.device_info.device_id = source_type;
-    event.base_event.device_info.audio_type = AUDIO_DEVICE_NONE;
+    event.base_event.device_info.audio_type = AUDIO_DEVICE_IN_HDMI;
     event.base_event.device_info.audio_address = NULL;
     event.base_event.type = event_type;
     switch (source_type) {
```

此种方式延时可能会小一些，但是录屏无法录制tvinput预览声音，用户可根据需求使用。